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RESEARCH ARTICLE

A Dual-Stream Deep Learning Architecture With Adaptive Random Vector Functional Link for Multi-Center Ischemic Stroke Classification

MAHESH ANIL INAMDAR[1], ANJAN GUDIGAR[2,3,4], U. RAGHAVENDRA[2], MASSIMO SALVI[5], RAJA RIZAL AZMAN BIN RAJA AMAN[6], NADIA FAREEDA MUHAMMAD GOWDH[6], IZZAH AMIRAH BINTI MOHD AHIR[6], MOHD SALAHUDDIN BIN KAMARUDDIN[6], KHAIRUL AZMI ABDUL KADIR[6], FILIPPO MOLINARI[5], (Senior Member, IEEE), AJAY HEGDE[7], GIRISH R. MENON[8], AND U. RAJENDRA ACHARYA[9,10]

[1]Department of Mechatronics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, India
[2]Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, India
[3]School of Medicine, Faculty of Health and Medical Sciences, Taylor's University, Subang Jaya 47500, Malaysia
[4]Digital Health and Medical Advancement Impact Lab, Taylor's University, Subang Jaya 47500, Malaysia
[5]Biolab, PolitoBIOMed Laboratory, Department of Electronics and Telecommunications, Politecnico di Torino, 10129 Turin, Italy
[6]Department of Biomedical Imaging, Universiti Malaya Medical Centre, University of Malaya Research Imaging Centre, Faculty of Medicine, Universiti Malaya, Kuala Lumpur 50603, Malaysia
[7]Manipal Hospitals, Bengaluru 560017, India
[8]Department of Neurosurgery, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, Manipal 576104, India
[9]School of Mathematics, Physics, and Computing, University of Southern Queensland, Springfield, QLD 4300, Australia
[10]Centre for Health Research, University of Southern Queensland, Springfield, QLD 4300, Australia

Corresponding author: Anjan Gudigar (anjan.gudigar@manipal.edu)

ABSTRACT One of the main causes of death and permanent disability is ischemic stroke, for which prompt and precise diagnosis is essential to successful treatment. This study introduces a novel dual-stream deep learning framework for ischemic stroke classification using Computed Tomography (CT) images, specifically addressing challenges in accuracy, computational efficiency, and clinical interpretability. Three significant innovations are included in the suggested architecture: (1) a hybrid Dual Attention Mechanism that combines Dynamic Routing and Cross-Attention for improved region-specific feature discrimination; (2) a Multi-Scale Feature Extraction Module with parallel convolutional pathways that captures both contextual and fine-grained features; and (3) an Adaptive Random Vector Functional Link layer that significantly reduces training time while maintaining high classification performance. When tested on a single-center dataset, the model achieves state-of-the-art classification accuracy of 98.83% across normal, acute and chronic stroke categories. We demonstrate the strong generalization capabilities of the proposed framework by achieving 92.42% accuracy on a diverse, multi-center dataset of 7,842 CT images. The integration of explainable Artificial Intelligence tools improve clinical trustworthiness by offering clear insight into the model's decision-making process. These outcomes demonstrate the model's potential to use in actual clinical settings for quick and accurate stroke diagnosis, along with its interpretability and computational efficiency.

INDEX TERMS Ischemic stroke detection, deep learning, cross-attention, random vector functional link networks, medical image analysis, explainable AI.

I. INTRODUCTION

According to 2020 estimates, there were 11.71 million stroke cases worldwide, with ischemic strokes accounting for approximately 65%, while Intracerebral Hemorrhage (ICH)

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and Subarachnoid Hemorrhage (SAH) comprised around 29% and 6%, respectively [1]. The world health organization reports that annually, 15 million people experience a stroke, of which 5 million are left with lasting disabilities, and more than 6 million result in death [2]. The impact is particularly significant in low and middle-income countries [3], with a lifetime risk of 24.9% in people above 25 years [4] with 10-20% of ischemic strokes being misdiagnosed in emergency settings [5], and global costs projected to rise from $891 billion in 2018 to $1.68 trillion by 2050 [6]. Hence, there is a critical need for more accurate and efficient diagnostic tools. Ischemic stroke results from an obstruction that impedes blood flow to the brain, depriving it of oxygen and nutrients, which can lead to cell death [7]. Brain cells are highly sensitive to hypoxia, and the extent of damage depends on ischemia and the adequacy of collateral blood supply. Urgent interventions, such as thrombolysis or mechanical thrombectomy, are employed to preserve viable brain tissue; however, their success depends upon multiple factors [8]. Figure 1 shows Computed Tomography (CT) brain scans images of normal case, acute cases (*i.e.,* brain tissue less than or 1 week after the onset of stroke) and chronic case (*i.e.,* brain tissue after 3 weeks after the onset of stroke). There are two main types of ischemic stroke, classified by the cause of the brain region's loss of blood flow-thrombotic and embolic strokes. Thrombotic strokes occur due to thrombus formation within intracranial arteries, typically associated with atherosclerosis, and can present with varying severity depending on the rapidity and extent of thrombus formation following plaque rupture, as well as the degree of arterial occlusion and collateral circulation adequacy. Embolic strokes are caused by clots formed elsewhere, commonly from atrial fibrillation, and typically present more suddenly than thrombotic strokes [9], [10]. CT and Magnetic Resonance Imaging (MRI) remain the primary imaging modalities in diagnosing stroke. Non-contrast CT (NCCT) is the first-line imaging tool due to its availability and speed, despite limited sensitivity in detecting early ischemic changes [11]. While MRI, more specifically diffusion-weighted imaging, shows better sensitivity in diagnosing early infarction [12], [13], its emergency use is limited by cost and availability. NCCT remains valuable for determining infarct size and location, especially in patients with MRI contraindications [14], [15].CT Angiography (CTA) provides excellent visualization of vascular anatomy, with sensitivity of 92-100% and specificity of 82-100% for large vessel occlusion detection [10], [14]. CT Perfusion (CTP) also provides collateral information about cerebral blood flow and penumbra extent [10]. The latest advancement, Dual-Layer Spectral CT (DLSCT), offers higher resolution imaging at reduced radiation doses, potentially replacing conventional NCCT in certain applications [14]. Machine Learning (ML) and Deep Learning (DL) have enhanced CT imaging capabilities through automated lesion detection and classification [15], [16]. Early detection of ischemic stroke is critical in improving clinical outcomes, with current guidelines emphasizing the "golden hour," where administering clot-dissolving treatments within 4.5 hours of symptom onset can significantly reduce long-term disability [17]. The need for automated systems is driven by frequent missed or delayed diagnoses of subtle early ischemic signs [18]. However, implementation faces several challenges: inconsistent imaging data, the need for interpretable results [19], [20], and limitations in current ML algorithms regarding overfitting and dataset availability [21]. Despite these challenges, such systems have the potential to expedite diagnosis and reduce human error.

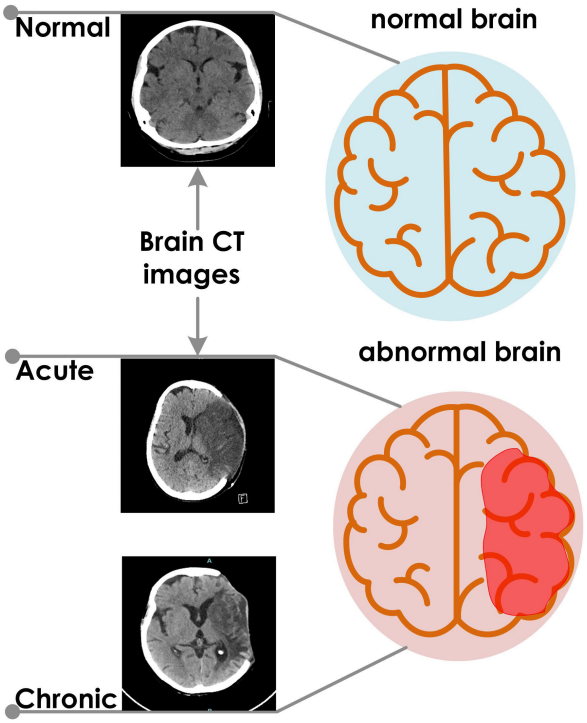


FIGURE 1. Brain scans displaying the two stroke cases.

A. MOTIVATION

In emergency medicine, where treatment choices must be made within a limited therapeutic window. An accurate and rapid detection of ischemic strokes from CT images is still a major challenge. Existing DL methods have demonstrated promise, but because brain CT interpretation is inherently complex, these methods frequently misclassify the cases and struggle with the subtle radiological markers that distinguish different stroke severities. Furthermore, the practical implementation of current models in clinical settings with limited resources is limited by their frequent high computational resource requirements. Effective, clinically applicable DL solutions that preserve accuracy and computational efficiency are needed.

B. CONTRIBUTIONS

- We propose a novel hierarchical multi-scale DL architecture with residual learning and dense connectivity

patterns, achieving maximum classification accuracy of 98.83% across three classes.
- We introduce an innovative hybrid attention mechanism that combines Dynamic Routing Attention and Cross-Attention in a dual-stream framework. This mechanism adaptively modulates feature importance through learnable routing coefficients while simultaneously capturing long-range dependencies across spatial dimensions, enabling precise localization of subtle stroke indicators with a significant improvement in region-specific feature extraction.
- We develop a computationally efficient framework by integrating Random Vector Functional Link (RVFL) networks with our DL pipeline, making the training faster while maintaining comparable performance.
- We enhance model interpretability through a comprehensive Explainable AI (XAI) framework that integrates multiple explanation methods. Our approach generates class-discriminative visual explanations of the obtained results.
- To the best of our knowledge, this is the first work on multi-center data, making it more resilient, robust and yields much more generalized predictions.

The paper is structured as follows: Section II reviews existing work across various categories. Section III details methodology, including data acquisition and experimental setup. Section IV presents results, comparisons, and ablation studies. Sections V and VI provide discussion and conclusions.

II. RELATED WORKS

Recent advancements in machine intelligence techniques have significantly enhanced the detection and classification of ischemic brain strokes using CT images [20], [21]. The more comprehensive review of the latest studies on brain strokes and their difficulty and challenges are presented in [22]. Here we group recent works (2020-2024) according to their approach.

A. FEATURE EXTRACTION TECHNIQUES WITH MACHINE LEARNING CLASSIFIERS

Traditional ML approaches continue to evolve, incorporating sophisticated feature extraction methods for enhanced stroke detection. Gautam and Raman proposed a Local Gradient of Gradient Pattern (LG2P) as a feature descriptor for classifying brain CT images (hemorrhagic stroke, ischemic stroke, and normal) [23]. The same author introduced in [24] a feature extractor based on Local Neighborhood Pattern (LNP) for stroke classification in CT images, utilizing quantum information concepts for enhanced classification.

B. SEGMENTATION

Recent segmentation approaches focus on developing specialized architectures for the precise delineation of stroke regions across various imaging modalities. U-Net-based architectures have shown promising results, with several adaptations: Kaya and Önal [25] employed transfer learning techniques for ischemic and hemorrhagic stroke segmentation, Li et al. [26] developed a multi-scale variant for enhanced lesion identification, and Korra et al. [27] integrated data augmentation for improved detection. Advanced architectural modifications include the MultiRes U-Net presented by Soltanpour et al. [28] that was able to handle various scales and irregular shapes. Makela et al. evaluates a CNN-based segmentation model using CTA data, which avoids radiation-heavy CTP scans and provides segmentation for suspected strokes [29]. Cross-modal approaches have demonstrated significant progress, as demonstrated by the work of Shi et al. [30], which uses cross-modal cross-attention to segment acute ischemic stroke lesions from multiple CTP maps. Kuang et al. [31] presented a new Convolutional Neural Network (CNN), I2PC-Net, to segment ischemic penumbra and infarct core regions in NCCT images of acute ischemic stroke patients. Kalyanasundaram et al. [32] proposed a predictive modeling for stroke subtype classification using ML techniques like Recursive Feature Elimination with Cross-Validation (RFECV) and extra trees classifier. Object detection models like YOLOv5 have been also integrated into stroke detection systems due to their real-time processing capabilities [21]. Following detection, models such as MedSAM can be employed for precise lesion segmentation, further enhancing diagnostic accuracy [21]. Jazzar and Douik proposes a hybrid encoder–decoder architecture that combines ResUNet++ and a modified U-Net with an atrous spatial pyramidal pooling block to enhance the stroke lesion segmentation and achieved a Dice coefficient of 0.85 [33]. Geetha et al. have used DeepLab-V3Plus and U-Net architectures to segment acute ischemic stroke lesions from diffusion-weighted MRI scans. They observed that DeepLab-V3Plus achieving superior segmentation performance when compared to U-Net [34].

C. DEEP LEARNING TECHNIQUES

Modern DL approaches leverage advanced architectures and training strategies to improve stroke classification. Gahiwal et al. used a CNN model to classify strokes in CT images, achieving a 90% accuracy using convolutional and pooling layers for feature extraction [35]. Neeti et al. presented a 3D CNN model for stroke classification in NCCT images, enhancing classification through custom preprocessing and addressing class imbalance with strided slicing [36]. Saleem et al. proposed a diagnostic system for early stroke detection using a combination of genetic algorithms and Bidirectional Long Short-Term Memory (BiLSTM) models, enhancing CT image classification and feature selection [37]. UmaMaheswaran et al. introduced a hybrid ML based on XGBoost for acute stroke detection in enhanced NCCT images [38]. Previous studies have shown that some researchers used this procedure to automatically extract relevant features from pretrained models and apply

a classifier [39], [40], [41]. Abulfaraj et al. introduced a feature fusion-based model for stroke identification using CT images, employing DL architectures (SqueezeNet v1.1 and MobileNet V3) combined with a CatBoost model to enhance stroke detection accuracy [42]. Fang et al. compared different DL (CNN, LSTM, ResNet) and ML (Support Vector Machine (SVM), Random Forest (RF), Deep Forest) methods for predicting 6-month outcomes of ischemic stroke using structured clinical data from the IST dataset, assessing the performance of deep forest methods on large medical datasets [43]. Raj et al. proposed a hybrid CNN and Vision Transformer (ViT) model named StrokeViT, combined with AutoML, to classify stroke types in CT images and predict outcomes at both slice and patient levels [44]. Qasrawi et al. introduced a Hybrid Ensemble DL (HEDL) model for ischemic stroke detection and classification, utilizing DenseNet121 and an enhanced Random Vector Functional Link (edRVFL) for improved accuracy in diverse stroke stages [21]. Sabir et al. proposed a novel deep CNN for early stroke detection in CT images within 6 hours of onset, focusing on rapid intervention capabilities. The model incorporates a feature extractor, fusion module, and stroke detection components [45]. Diwan et al. presented a deep dense neural network model for early stroke detection using patient medical data, achieving an accuracy of 95% on binary classification of stroke prediction [46]. Recently, Inamdar et al. have proposed a self-supervised technique for stroke classification and achieved better results [47]. Dong et al. proposed the self-supervised learning model for brain CTA to address multiclass problems, such as detection of acute ischemic stroke, classification of intracerebral hemorrhage, and prediction of ischemic core volume [48]. This model leverages contrastive learning using 3D CTA images and radiology reports and thus demonstrating improved generalizability across these tasks. Another multiclass classification approach is proposed by Hossain et al. where in ViT and LSTM networks are integrated to detect and classify strokes using CT scans [49]. The method outperformed CNN-based approaches by achieving an accuracy of 96.61%. In contrast, Amador et al proposed a cross-attention based DL model, which focuses on a binary classification task and predicts functional stroke outcomes using 4D CTP imaging and clinical metadata [50].

D. MULTI-MODAL TECHNIQUES

Multi-modal approaches combine different data sources and imaging modalities to enhance stroke detection and outcome prediction. Samak et al. proposed a transformer-based multimodal model that combined clinical data and NCCT imaging to predict stroke treatment outcomes, showing high performance on the modified Rankin Scale (mRS) [51]. Raju et al. [52] investigated multimodal fusion strategies (early, late, bottleneck, and hierarchical) for stroke lesion segmentation using CT perfusion data. Borsos et al. used multimodal DL to predict post-stroke functional outcomes by combining CT perfusion and tabular clinical data. They introduced a Dynamic Affine Feature Map Transform (DAFT) and evaluate various data fusion techniques [53]. The proposed multimodal model produced an Area Under Curve (AUC) of 0.75 and F1 score of 80% with superior prediction over baseline models for stroke outcome prediction.

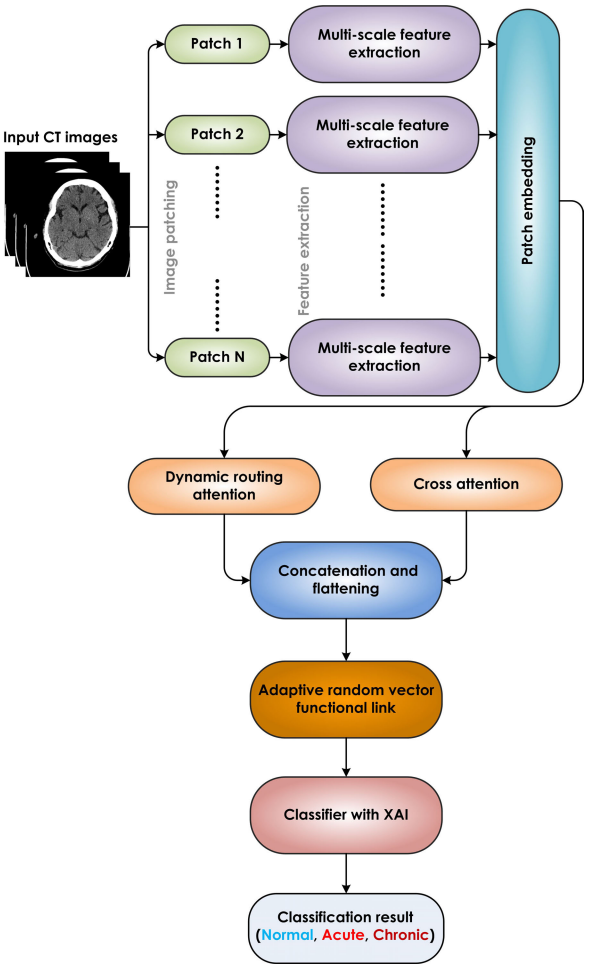


FIGURE 2. Overview of the proposed architecture.

E. LIMITATIONS IN EXISTING APPROACHES

While recent studies have advanced stroke detection significantly, several limitations persist in current approaches. Several studies [25], [26] make use of architecture such as U-Net in achieving segmentation tasks, focusing majorly on stroke detection using single-scale or limited multi-scale features. Though these methods prove adequate for segmentation tasks, they fail to incorporate contextual features necessary for accurate stroke classification. In this work, we adopt a Multi-Scale Feature Extraction Module that ensures simultaneous processing of local and global patterns,

by having parallel convolutional pathways that would employ kernels of varying sizes. Richer feature representations along with an increase in sensitivity toward subtle ischemic patterns would enable more robust classification results compared to single-scale methods.

Attention mechanisms have been used extensively in stroke detection, and one recent example is C2MA-Net by [30], incorporating cross-modal cross-attention for segmentation. Although well-performing on cross-modal tasks, these models often ignore the adaptive weighting of spatial features under a single imaging modality. The dual attention mechanism proposed here, which integrates Dynamic Routing with Cross Attention, shows superior feature discrimination. The realizability of our methodology to spatial dependencies with important regions culminates in an increase in region-specific feature extraction performance, when compared with attention-based methods such as those suggested by [45]. Many studies [36], [52] validated their models on single-center data sets, thereby restricting their generalizability to different clinical settings. On the other hand, our model validation on a multi-center dataset of 7,842 CT images shows the model's strength and scalability. The ability to achieve 98.83% (maximum) accuracy for single-center dataset across three classes (*i.e.,* normal, acute and chronic) highlights its resilience to dataset variability, outperforming single-center models like the 3D CNN [36] (92% accuracy). Explainability is an important issue that is sometimes ignored in stroke detection research. Models such as those by [45] provided limited interpretability, which hampers clinical trust. Our model incorporates advanced explainability tools like SHapley Additive exPlanations (SHAP), offering both visual and quantitative insights into feature importance. This helps clinicians to interpret the decision-making of predictions, thus enhancing the use in medical practice and overcoming the ''black box'' aspects of many DL models. Finally, computationally demanding techniques, e.g., hybrid CNN-ViT proposed in [44], pose challenges for deployment in resource-limited settings. Our model makes a trade-off by combining simple and efficient mechanisms Adaptive RVFL (ARVFL), as well as a simple multi-scale architecture. This allows for good performance on both high-end and low computational resource scenarios. Table 1 shows the summary of recent works (2020-2024) according to their approach.

III. PROPOSED METHODOLOGY

In this paper, we develop a DL-based classification model for ischemic brain scans using a structured CNN-based encoder for feature extraction. The architecture comprises three key innovative components: a multi-scale feature extractor, a dual attention mechanism, and an adaptive classification layer. The encoder progressively increases filter sizes *(i.e., 32, 64, 128 and 256)* and utilizes multiple kernel sizes (*i.e., 3*, *5*, and hybrid combinations) to capture multi-scale spatial characteristics. Features are processed through parallel Dynamic Routing (DR) and Cross-Attention (CA) mechanisms, which enable adaptive weighting of spatial regions. Further, an ARVFL layer with adaptive parameters (*i.e., 1,2,4*) is utilized. Finally, the extracted features from the model are classified using top ML models. In addition, XAI is adapted to understand feature interpretability of the proposed model. In Figure 2 we present the complete architecture of the proposed approach.

A. DATA ACQUISITION

Our study utilizes a multi-center dataset comprising 7,842 CT images collected between 2023-2024 from two medical centres, Department of biomedical imaging, Faculty of Medicine, Universiti Malaya Medical Centre (UMMC), Malaysia and Department of Neurosurgery, Kasturba Medical College (KMC), India. The dataset includes 2,100 acute cases, 1,347 chronic cases, and 4,395 normal cases (Refer to Table 2). Institutional ethical committee permission is obtained prior to the study. Figure 3 shows the examples of the sample images of each case collected from two medical centres. All images were standardized to 512 $\times$ 512 $\times$ 3 pixels. The data was split randomly 10 times with 90% for training and 10% for testing, maintaining consistent class distributions. The average performance is taken to show the efficacy of the proposed system. Implementation was carried out using Python 3.9 with Keras/TensorFlow on an NVIDIA GeForce RTX 3070 GPU system.

B. ARCHITECTURE COMPONENTS

The details of architecture of the proposed model are given in the following sections.

1) MULTI-SCALE FEATURE EXTRACTION MODULE

Our novel multi-scale feature extraction network processes input images through parallel convolutional pathways with varying kernel sizes (*i.e., 3 $\times$ 3*, *5 $\times$ 5*, and combination of *3 $\times$ 3* and *5 $\times$ 5*). Each pathway implements a progressive channel dimension *(32$\rightarrow$64$\rightarrow$128$\rightarrow$256)* with interspersed max-pooling operations (refer to Figure 4). Initially, input brain CT image is divided into four non-overlapping patches and each patches are given to proposed multi-scale feature extractor unit. Further, these features are concatenated and performed global average pooling to generate feature maps for each corresponding patch. This design enables simultaneous capture of fine-grained details and broader contextual features, creating rich multi-resolution representations particularly suited for stroke detection. The complete feature extraction unit is shown in Figure 4 and it is given as

$$\mathbf{F} = E\{\mathbf{I}\} \tag{1}$$

where,

- $E\{.\}$ is the multi-scale operator.
- I input image

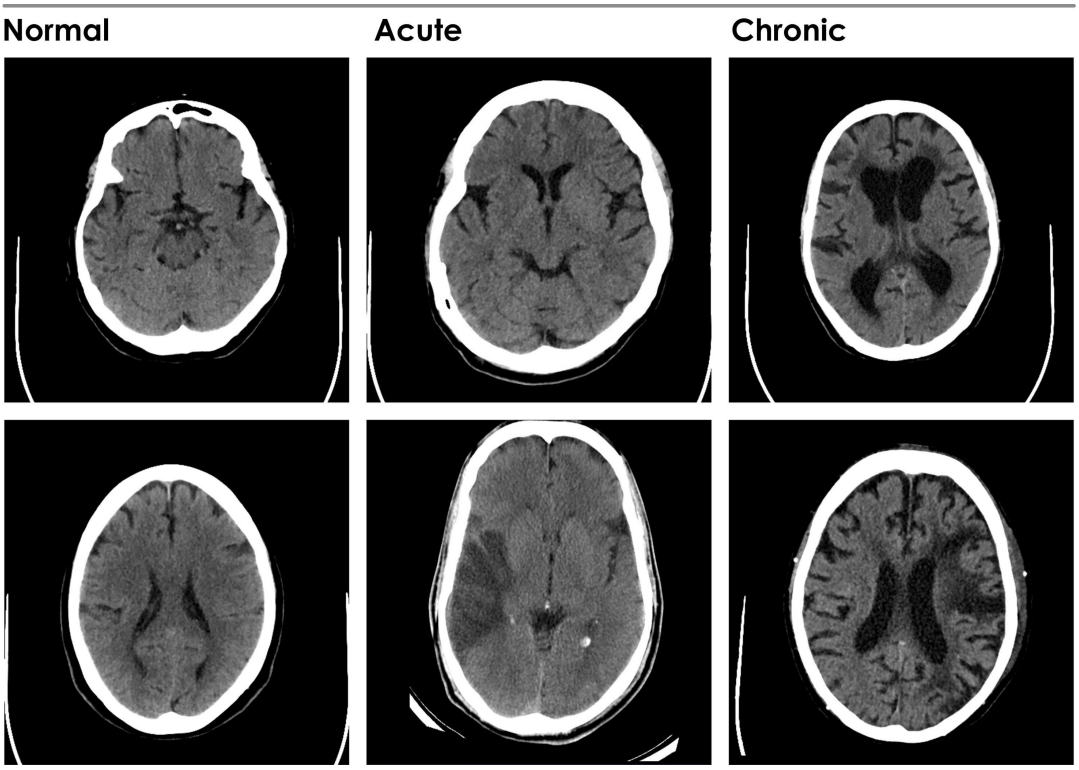


FIGURE 3. Samples Images used in the current study.

- $F \in \mathbb{R}^{N \times D}$ extracted feature tensor, N is the number of patches(4), D is the dimensionality of the embeddings (256).

Finally, (4×256) patch embeddings are generated for each brain CT image and further it is given to DA module to highlight the significant features by computing attention scores.

2) DUAL ATTENTION MODULES

The dual attention mechanism combines DR and CA to enhance feature discrimination.

Dynamic Routing is a central mechanism first introduced by Sabour et al. [54] addresses limitations in CNNs by learning spatial hierarchies through feature hierarchies. This mechanism is particularly valuable in applications requiring correct part-whole relationships, such as image recognition and segmentation [55], [56]. The DR algorithm determines the probability of each capsule's output matching with higher-level capsules. The process involves:

a. Routing Coefficient Initialization:

$$d_{xy} = 0, \quad \forall \hat{i}, \hat{j} \tag{2}$$

where: d_{xy} Logits for routing coefficients between the $\hat{i}$-th input and $\hat{j}$-th output capsule.

b. Coupling Coefficients:

$$p_{xy} = \frac{\exp(d_{xy})}{\sum_z \exp(d_{xz})} \tag{3}$$

where p_{xy} Normalized coupling coefficient (softmax of d_{xz}).

c. Output Capsule:

$$\mathbf{w}_y = \text{Squash}\left(\sum_x p_{xy}\, \mathbf{t}_{x|y}\right) \tag{4}$$

where,

- $t_{x|y} = W_{xy}, t_x$: The transformed input capsule using weight matrix W_{xy}
- t_x is the x-th input capsule.
- w_y Output capsule vector.
- $t_{x|y}$ Transformed input capsule.

d. Logit Update

$$d_{xy} \leftarrow d_{xy} + \mathbf{t}_{x|y} \cdot \mathbf{w}_y \tag{5}$$

Cross Attention Mechanism popularized by Vaswani [57], enables learning relationships between different sequences or feature sets. It has proven effective in multimodal and multi-source tasks [58], [59], [60], [61]. The attention mechanism computes a weighted representation of the values based on the similarity between the queries and keys. This can be formulated as:

a. Attention Scores: Attention weight between query i and key j, *i.e.,* γ_{ij} is computed using softmax function.

b. Weighted Summation:

$$\mathbf{Y}_i = \sum_j \gamma_{ij}\, \mathbf{c}_j \tag{6}$$

TABLE 1. Comparison of literature review on automated ischemic brain stroke using CT images.

Paper & Year	Dataset Details	Merits	Remarks
Kaya et al. 2023 [25]	CT images: 4,000	High segmentation with Prec.: 95.06%.	Relies heavily on pre-trained models, limiting adaptability across varying datasets.
Li et al. 2021 [26]	Stroke patients: 30	High segmentation precision with a Dice similarity coefficient (DSC: 86%).	U-Net's view limitations may affect segmentation accuracy at certain angles (Acc:95.8%).
Korra et al. 2024 [27]	–	Achieves high accuracy (Acc.: 94.57% (2-class)) in stroke detection.	Relies on large amounts of data augmentation, which may not generalize well to all real-world datasets.
Soltanpour et al. 2021 [28]	ISLES challenge 2018 database	Improved segmentation accuracy (DSC: 68%), especially with varied lesion appearances and scales.	Requires significant preprocessing, including contralateral imaging and Tmax heatmaps, which may complicate deployment.
Makela et al. 2022 [29]	Patients: 150	Reduced radiation exposure compared to CTP and quicker integration into existing CTA protocols. Sen.: 0.54, Prec.: 0.69, Acc.: 88%.	Limited precision, particularly in small or ambiguous lesions, leading to moderate sensitivity in clinical applications.
Shi et al. 2021 [30]	Training: 94, Testing: 62 cases	Better segmentation results (ISLES-2018, Test: Rec.: 0.59) due to cross-modal attention, which captures complex relationships between modalities.	Complexity of the model could lead to higher computational costs and training times.
Kuang et al. 2024 [31]	NCCT Scans: 197	Uses only NCCT, making it faster and more accessible than advanced imaging like CT perfusion. DSC: 65.67%.	Lower accuracy compared to CT perfusion methods in distinguishing between ischemic penumbra and infarct core regions.
Gahiwad et al. 2023 [35]	CT images: 2,551	High accuracy (Acc.: 90%) for early stroke detection.	Requires a large dataset and substantial computational resources for optimal training.
Neethi et al. 2022 [36]	Patient: 234	High F1-score improvement, demonstrating superior stroke classification accuracy (Acc.: 90%).	Computationally intensive due to 3D processing requirements.
Saleem et al. 2024 [37]	Normal: 1,551, Stroke: 950 Images (Kaggle)	Achieves a high classification accuracy (Acc.: 96.5% (2-class)).	BiLSTM and genetic algorithms are computationally intensive, which may limit scalability.
UmaMaheswaran et al. 2024 [38]	Normal images: 1551, Stroke images: 950	Provides high accuracy (Acc.: 97%) and low false positive rate.	Requires extensive preprocessing, making it time-consuming.
Patel et al. 2023 [39]	CT images: 100	Lightweight and suitable for real-time applications on limited-resource devices.	May lack accuracy in edge cases due to its compact structure (Acc:97%).
Diker et al. 2023 [40]	CT images: 2,501	High classification accuracy (Acc.: 97.06%), suitable for accurate stroke detection.	Computationally intensive, challenging for low-resource settings.
Irfan et al. 2024 [41]	CT images: 2,501	High feature reuse through dense connections, yielding high classification accuracy (DenseNet-169).	Higher computational demand, making it less efficient for mobile platforms.
Abulfaraj et al. 2024 [42]	Normal: 1551 & Abnormal: 950 images	Achieves high accuracy (99.1%), Prec.: 99%, Rec.: 98% due to feature fusion, enhancing the model's ability to capture complex patterns.	The complex model may require more computational resources, potentially limiting deployment in resource-limited settings.
Fang et al. 2022 [43]	Patients: 16,403	Demonstrates Deep Forest's effectiveness (Acc.: 83% (for CNN and RF)) with lower computational costs than deep learning.	Shows limited improvement in predictive accuracy compared to ML.
Raj et al. 2023 [44]	–	Achieves high patient-level accuracy (92%) using AutoML, reducing radiologist workload.	High resource demands for both CNN and ViT integration.
Zeynal et al. 2023 [51]	–	Achieves high AUC (0.85), leveraging both imaging and clinical data.	Requires large datasets and high computational power due to the transformer architecture.

DSC: Dice Score; Acc: Accuracy; Sens: Sensitivity; Prec: Precision; Rec: Recall.

- $c_j \in R^d$: Value vector corresponding to key j.
- Y_i: Output vector for the i-th query after applying attention.

The enriched (weighted feature vectors with attention scores) features from DR and CA modules F_{fused} is then fused and fed into ARVFL (classification head) unit, where F_{fused} is given

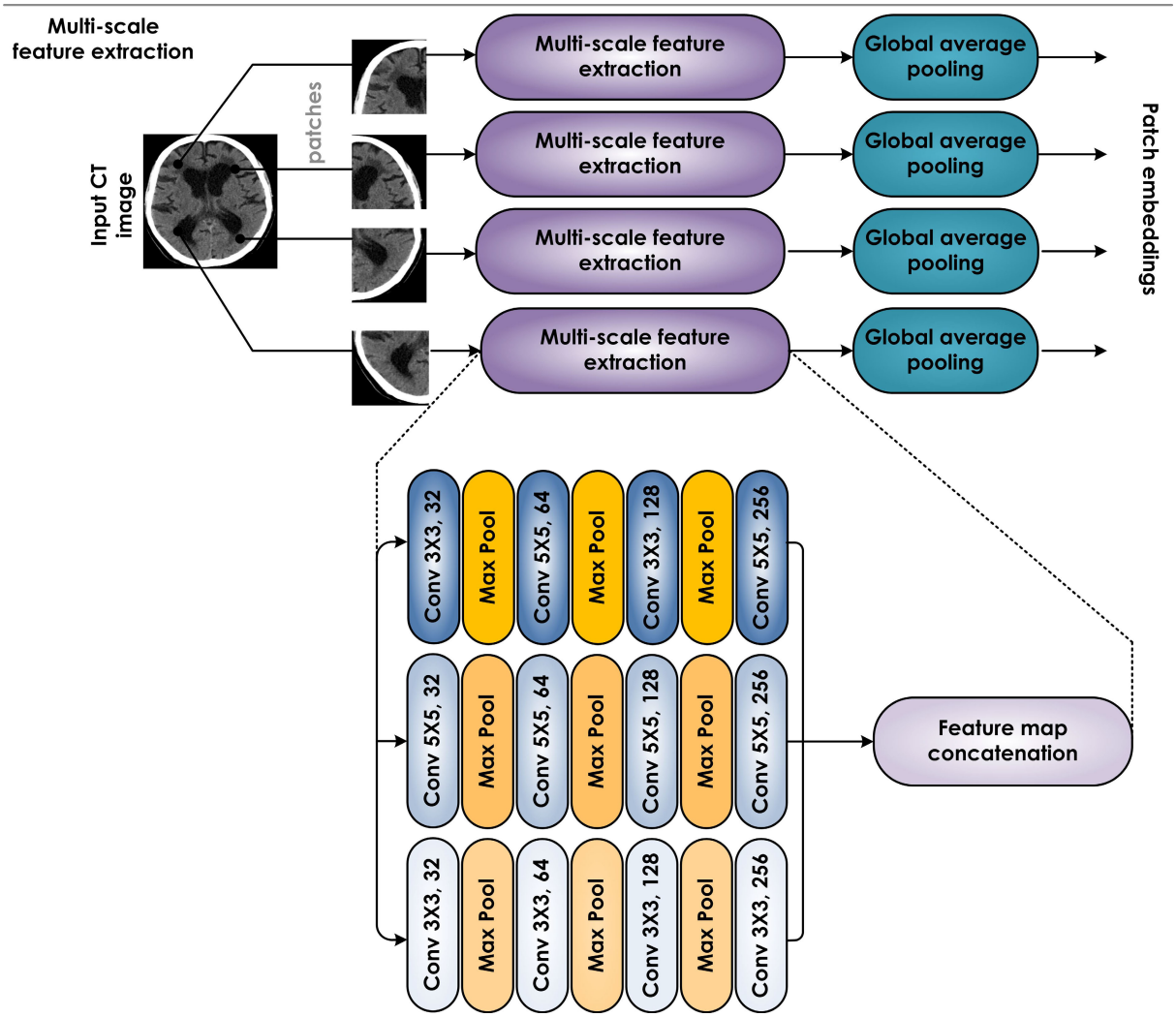


FIGURE 4. Architecture of multi-scale feature extractor.

TABLE 2. Dataset details used in the current study.

Dataset	Normal	Acute	Chronic
Dataset 1 (UMMC, Malaysia)	3,451	742	505
Dataset 2 (KMC, India)	944	1,358	842

by

$$\mathbf{F}_{\text{fused}} = \text{Flatten}\{\text{Concatenate}(\text{DR}(\mathbf{I}), \text{CA}(\mathbf{I}))\} \quad (7)$$

3) ADAPTIVE RANDOM VECTOR FUNCTIONAL LINK

RVFL was first introduced by Huang et al. [62] and represents a feedforward neural network with random weights and direct input-output links. This design simplifies training while capturing complex patterns without backpropagation [63], [64], [65]. ARVFL layers improve upon traditional RVFL networks by incorporating trainable factors that dynamically modulate randomly initialized weights. This adaptability allows the model to efficiently learn intricate patterns in data, bridging the gap between simplicity and expressiveness while maintaining computational efficiency. Figure 5 provide a schematic representation of ARVFL.

The steps for ARVFL are as follows:

1) Weight generation: The weight generation phase includes the initialization of random weights and the adaptive modulation step and is given by,

$$\mathbf{W}_{\text{mod}} = \mathbf{W}_{\text{rand}} \odot \mathbf{A}_{\text{adapt}} \quad (8)$$

where,

- $\odot$ Elementwise multiplication,
- $W_{\text{rand}} \in \mathbb{R}^{d \times u}$ Randomly initialized, fixed weights,
- $A_{\text{adapt}} \in \mathbb{R}^{d \times u}$ Trainable adaptive factors,
- $W_{\text{mod}} \in \mathbb{R}^{d \times u}$ Modulated weights used for feature transformations, and
- u and d is the number of neurons and is the input dimensionality respectively.

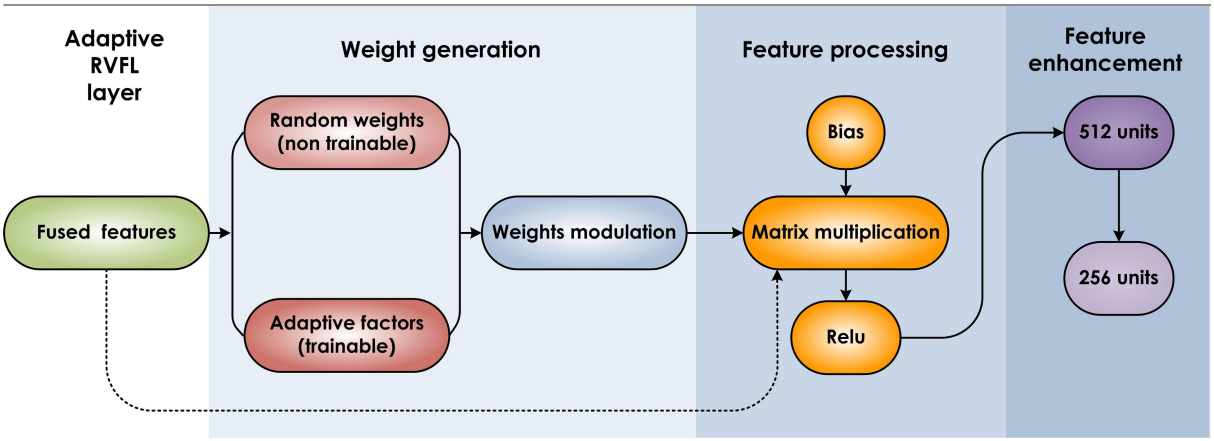


FIGURE 5. Schematic representation of adaptive RVFL layer.

Here W_{rand} brings randomness, ensuring diversity in feature mapping, while A_{adapt} adapts the weights during training, introducing flexibility to optimize the network's performance dynamically.

2) Feature Processing: It is performed by,

$$\mathbf{H} = \mathbf{X} \cdot \mathbf{W}_{\text{mod}} + \mathbf{B} \tag{9}$$

where,

- $X \in \mathbb{R}^{n \times d}$ Input feature matrix with n samples,
- $W_{\text{mod}} \in \mathbb{R}^{d \times u}$ Modulated weight matrix,
- $B_{\text{mod}} \in \mathbb{R}^{n \times u}$ Trainable bias vector, and
- $H \in \mathbb{R}^{n \times u}$ Transformed feature matrix.

The matrix multiplication between X and W_{mod} captures linear relationships, while the bias vector B shifts the output to explore additional regions in the feature space.

3) Feature Enhancement: It is performed using following equation

$$\mathbf{H}_{\text{enhanced}} = f_{\text{activation}}(\mathbf{H} \cdot \mathbf{W}_{\text{proj}}) \tag{10}$$

where,

- $H \in \mathbb{R}^{n \times u}$ Processed feature matrix,
- $W_{\text{proj}} \in \mathbb{R}^{u \times v}$ Projection matrix, where v is the number of units in the enhanced layer, and
- $H_{\text{enhanced}} \in \mathbb{R}^{n \times v}$ Final enhanced feature matrix.

Here ReLU is considered as the activation function. By projecting features to dimensions such as *512* or *256* units, the network gains the capacity to learn higher-order feature interactions. This step is crucial for downstream tasks requiring fine-grained representations.

The final output is passed through the classifiers to predict the class probabilities.

C. ML CLASSIFIERS

In this section we present the generic description of the three classifiers employed: Catboost, XGBoost, and Label Spreading.

CatBoost (Categorical Boosting, CatB) is a gradient-boosting algorithm introduced in [66]. It employs a novel permutation-driven technique called ordered boosting to reduce prediction shift and overfitting, which is particularly effective in scenarios with categorical and high-cardinality features. It utilizes gradient boosting, which involves minimizing the loss function by iteratively fitting decision trees to the residual errors of the model. The update rule for the score $f(x_k)$ in the c-th iteration is:

$$f_c(x_k) = f_{c-1}(x_k) + \eta \cdot h_c(x_k) \tag{11}$$

where,

- $f_{c-1}(x_k)$ is the score of the previous iteration,
- $h_c(x_k)$ is the output of the c-th decision tree, which is fitted to the residuals (or gradients) of the loss function, and
- η is the learning rate.

XGBoost (Extreme Gradient Boosting, XGB) enhances gradient boosting by employing a regularization term to control model complexity, making it robust against overfitting [67]. The regularization for XGBoost includes both the traditional loss L and a tree function $\Omega(f)$ is given by:

$$\zeta_\phi = \sum_a L(b_a, \hat{b_a}) + \sum_m \Omega(f_m) \tag{12}$$

where,

- b_a is the true label,
- $\hat{b}_a$ is the prediction
- m is the number of trees.

Label Spreading (LS) [68] is a semi-supervised learning technique that propagates label information from labeled to unlabeled data points based on graph-based relationships. The algorithm uses a Gaussian kernel to determine similarity between data points and iteratively updates label distributions.

D. EXPLAINABILITY FOR FEATURE INTERPRETABILITY

SHAP is a game-theoretic approach explaining the outputs of machine learning models by computing the contribution of features to predictions. Based on Shapley values from cooperative game theory, SHAP distributes "payouts" across

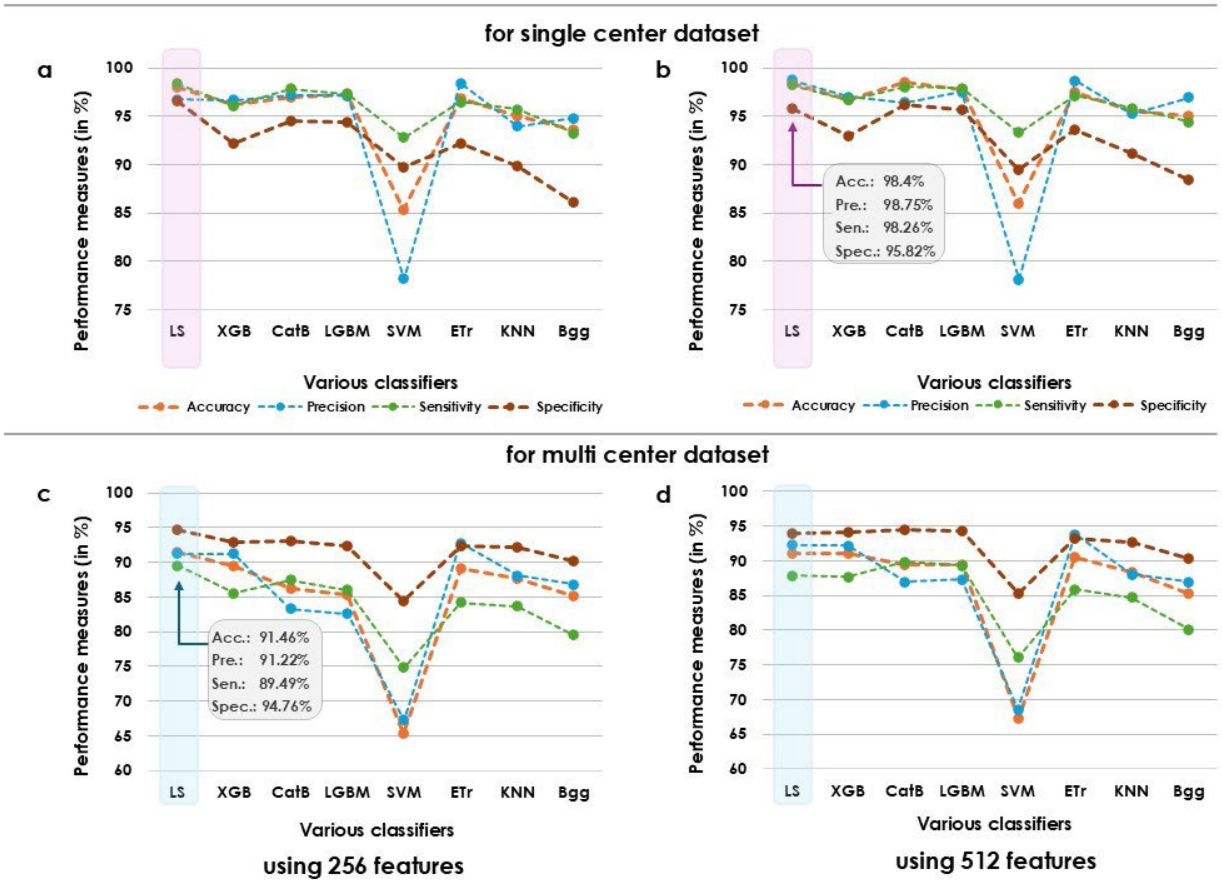


FIGURE 6. Comparative analysis of ML models on single-center (a,b) and multi-center (c,d) datasets using 256 features (a,c) and 512 features (b,d).

features, quantifying influence on model output [69]. SHAP extends further the model transparency, and its application areas in health care, finance, etc., are high-stakes decision-making areas where it needs to understand the reasoning behind the model. Three basic properties form the basis of the framework: consistency (proportional increase in Shapley values with feature importance), missingness (zero contribution for absent features), and local accuracy (accurate reconstruction of model predictions). These characteristics guarantee accurate and consistent interpretation of feature importance. There are several ways to implement SHAP, ranging from precise computation for small feature spaces to effective approximations like TreeSHAP and KernelSHAP, which provide flexibility in striking a balance between explanation precision and computational complexity. In medical applications, especially the diagnosis of stroke, SHAP helps physicians to:

- Find important biomarkers that influence predictions
- Verify model logic against domain expertise.
- Examine specific instances using local justifications.
- Make sure the decision is impartial and fair.

By offering interpretable, practical insights into model behaviour, feature contribution analysis's transparency promotes clinical trust and makes it easier to incorporate ML models into medical practice.

TABLE 3. Summary of performance (%) for single-center (*i.e.*, Dataset 1) data with 256 features.

Models	Accuracy	Precision	Sensitivity	Specificity	F1 Score
LS	97.78±0.44	96.17±0.94	96.54±1.04	98.18±0.60	96.26±0.80
XGB	96.87±0.60	96.84±0.66	93.83±1.35	96.47±1.53	95.04±0.74
CatB	97.27±0.37	95.60±1.39	96.31±0.93	97.68±1.10	95.77±0.63

TABLE 4. Summary of performance (%) for single-center data with 512 features.

Models	Accuracy	Precision	Sensitivity	Specificity	F1 Score
LS	97.89±0.49	97.46±1.00	96.02±1.20	97.78±0.79	96.57±0.84
XGB	97.15±0.47	97.28±0.84	94.36±1.17	96.68±1.34	95.55±0.74
CatB	97.51±0.46	95.61±0.82	96.63±0.80	97.95±0.66	96.00±0.44

TABLE 5. Summary of performance (%) for multi-center data with 256 features.

Models	Accuracy	Precision	Sensitivity	Specificity	F1 Score
LS	91.50±0.65	91.19±0.73	89.51±0.85	94.80±0.38	90.29±0.77
XGB	88.88±0.87	90.20±0.54	84.72±1.35	92.58±0.64	87.00±0.96
CatB	88.56±1.22	87.18±1.71	87.17±1.11	93.37±0.54	87.02±1.16

TABLE 6. Summary of performance (%) for multi-center data with 512 features.

Models	Accuracy	Precision	Sensitivity	Specificity	F1 Score
LS	90.66±0.66	91.63±0.81	87.84±1.01	93.91±0.44	89.51±0.88
XGB	90.40±0.44	91.85±0.57	86.33±0.88	93.57±0.26	88.65±0.63
CatB	90.48±0.32	89.91±0.47	88.19±1.00	94.22±0.32	88.94±0.52

IV. RESULTS

A. EVALUATION FRAMEWORK

Our model evaluation system employs comprehensive metrics to ensure thorough validation across multiple performance dimensions. Accuracy serves as the primary metric, measuring overall correctness through the proportion of true positives *(TP)* and true negatives *(TN)* to total outcomes, thus providing a general performance assessment [70]. Sensitivity, (or Recall) reflects the model's ability to identify true positives, calculated as $TP/(TP + FN)$, which proves crucial for detecting actual positive cases [71]. The specificity metric indicates the model's effectiveness in identifying true negatives, calculated as $TN/(TN + FP)$, and remains essential for minimizing false positives [72]. Precision assessment examines the proportion of true positives in positive predictions, calculated as $TP/(TP + FP)$, becoming particularly important when false positives are significant [70]. For imbalanced datasets, the F1 Score provides valuable insight by using precision and recall [70].

B. PERFORMANCE ANALYSIS OF PROPOSED ARCHITECTURE

Here we explain in brief the results, classifiers employed and the rationale of choosing these. The empirical analysis of our top-performing models reveals LS's excellent discriminative capabilities, achieving peak performance in single-center configurations with 98.83% accuracy, 98.32% precision, and 98.15% sensitivity and 98.23% F1 score at *256* features. This balanced achievement of *TP* and *TN* (specificity: 99.02%) demonstrates optimal decision boundary formation. When scaled to *512* features, the model maintains its robust accuracy (98.83%) while showing a marginal improvement in precision to 99.05%, suggesting enhanced capability in minimizing FP, though with a slight trade-off in sensitivity (97.42%) and achieving an profound 98.21% in F1 score. XGB and CatB demonstrate strong but consistently lower performance patterns. XGB achieves 97.77% accuracy in single-center *256* features, while CatB reaches 97.98% accuracy. The marginal differences in their TP rates (XGB 94.85% vs CatB 96.53% sensitivity) suggest varying approaches to handling decision boundaries, with CatB showing slightly better capability in identifying positive cases correctly. The model selection solely lies on their performance and the credibility in capturing importance features.

A critical observation emerges in multi-center scenarios, where all models experience performance degradation, though with varying degrees. LS maintains leadership with 92.42% accuracy, 91.89% precision, and 90.71% sensitivity at *256* features. The preservation of high specificity (95.36%) in multi-center data is particularly noteworthy, indicating robust control over false positives even in heterogeneous data distributions. The relatively smaller degradation in LS's performance (Δ accuracy $= -6.41\%$ from single to multi-center) compared to XGB (Δaccuracy $= -7.65\%$) and CatB (Δaccuracy $= -8.37\%$) suggests superior capability in handling center-specific variations.

Ten random partition average results, as shown in Tables 3 - 6, provide crucial insights into model stability. LS demonstrates remarkable consistency with accuracy standard deviations of $\pm 0.44\%$ in single-center features, significantly outperforming the stability metrics of gradient boosting approaches. The precision-sensitivity relationship in the 10-set experimentation analysis (96.17 $\pm 0.94\%$ precision, 96.54 $\pm 1.04\%$ sensitivity) indicates stable decision boundary formation across different data subsets, a critical factor for clinical applications.

Besides these main classifiers, we also explored Light Gradient Boosted Machines (LGBM) [73] and Extra Trees Classifier (ETc) [74], which handle high-dimensional, complex features well, crucial in medical imaging. SVM [75] and K-Nearest Neighbors (KNN) [76] are beneficial when class boundaries are clear and local feature patterns are important. Bagging (Bgg) stabilizes predictions across image feature variability, reducing overfitting, especially useful in domains with noisy datasets [77]. It is observed from the Figure 6 that LS outperformed when compared to other classifiers using a randomly selected set.

C. ABLATION STUDY

To better understand the contribution of each component in our architecture, we performed an ablation study by systematically removing key elements such as DR, CA, and the ARVFL layer. Initially, the full model, incorporating all these components, demonstrated superior performance across all metrics. However, when DR was removed, we observed a notable decrease in model accuracy and sensitivity, suggesting that the routing mechanism plays a crucial role in capturing the spatial hierarchies of the features and refining the classification process. The absence of DR resulted in a less precise aggregation of feature representations, which significantly impacted the model's ability to differentiate between complex patterns in the data. Next, removing the CA mechanism led to a increase in the model's precision and F1 score, highlighting the importance of feature alignment across different modalities or regions of the input data. However, the CA mechanism enhances the proposed model's ability to focus on discriminative parts of the image, improving the overall decision-making process. Lastly, when the RVFL layer was excluded, the model's performance showed a decrease in generalization ability, with a reduction in both F1 score and specificity. The RVFL layer, by projecting features into higher-dimensional spaces, aids in better classification boundaries, especially

in the presence of complex feature interactions. Overall, the ablation study underscores the significance of each architectural component, demonstrating that removing any one of these crucial modules negatively affects the model's capacity to effectively classify the data. Table 7 shows the results obtained for our ablation study.

D. COMPARISON IN PRETRAINED BACKBONE MODELS AS FEATURE EXTRACTOR

We compare the performance of our proposed model (presented in Table 8), that integrates a feature extractor with some state-of-the-art (SOTA) backbone (concise) architectures: MobileNet [78], DenseNet121 [79], and EfficientNetB0, EfficientNetB1 [80], NASNet [81], ConvNeXtBase [82], InceptionResNetV2 [83] and ViT [84]. The choice of models was based on their different complexities and efficiencies in extracting deep features from medical imaging data. Here, MobileNet, EfficientNetB0 and EfficientNetB1 and are lightweight architectures for good computational efficiency and thus suitable for deployment in resource-constrained environments. On the other hand, DenseNet121 have deeper, much more advanced architectures to be more capable of capturing complicated patterns in high-resolution medical images. Transfer learning indeed is very important in the medical area. It enables pre-trained models on large general datasets to be fine-tuned for specific medical tasks considerably enhancing model performance despite limited domain-specific data. This significantly reduces the need for huge, labelled datasets of medical data, which are often in scarce supply, and the approach improves the accuracy of models.

E. MODEL INTERPRETATION USING XAI

In Figures 7 & 8 we plot the decision plot and the summary plot. We first explain the summary plot which is in form of stacked histogram to provide a concise overview of feature importance and their influence on model predictions from the obtained SHAP values for the LS (best performing model) both for single and multi-center data with *256* and *512* features. The *y*-axis provides the list of top 10 most contributing features in descending order of their influence. The x-axis represents the mean absolute SHAP values for each feature, which quantify the average magnitude of the features contribution to the model's contribution. Larger the range indicate stronger average feature importance. These values are the raw values computed by the SHAP algorithm. The length of each feature bar (color coded to represent different classes) indicates it contributes more to the model's prediction on average across all class. The proportion of each segment within a bar indicates the contribution of that features to a particular class. For instance, for single centre dataset, in the case of *256* feature set, 218th feature has most contribution towards class 2, and has least saying for class 1. 189th feature on the other hand, biggest contribution equally towards class 2 and 1 (refer to Figure 8).

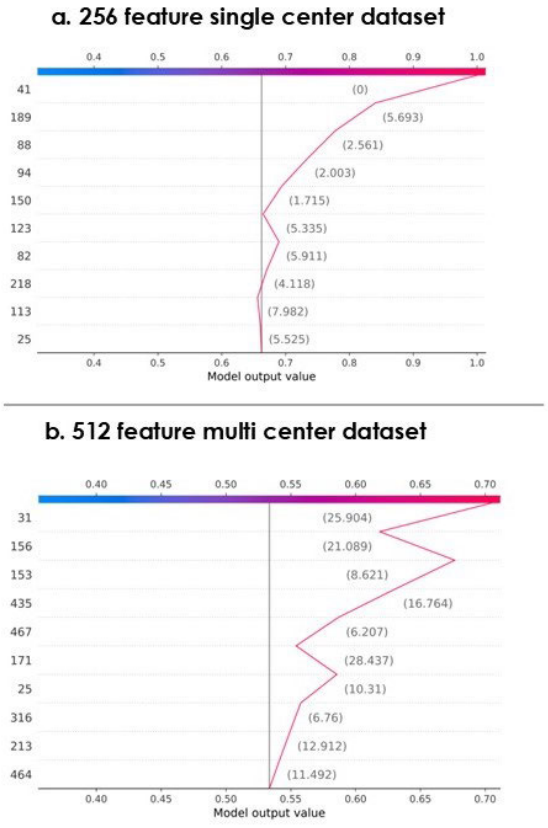


FIGURE 7. Decision plots of mean SHAP values for LS model for normal sample.

Similarly, for *512* features, the top feature 212, has equal and most contribution in case of class 0 and 1 and so on. Such similar explanation could be given in case of multi-centre dataset. An important observation is seen that all features are almost constant contributors for each class 0 which is the label for normal class, this is evident as normal CT scans share most features very similar to acute (label 1) and chronic (label 2). By observing the cumulative color distribution, it can infer whether certain classes dominate feature contributions across the model or whether the contributions are evenly distributed. This plot not only reveals which features are most important but also highlights their class specific effects offering a granular view of the model's decision-making process (refer to Figure 8).

To extend the interpretation of the obtained results in Figure 7 we showcase the decision plots for the top performing model (i.e., LS) with the corresponding set *i.e., 256* features for single centre data and *512* features for multi-centre data for a normal sample (class label 0). Here we see the same top 10 features as observed in summary plot but in varying proportion of their respective contributions for predictions (shown as logits). For the Figure 7 a) The top 10 feature contributions here shows a more distributed and balanced pattern, with smaller contribution extremes.

TABLE 7. Summary of performance (%) obtained for ablation study.

Model Configuration	Accuracy	Precision	Specificity	Sensitivity	F1 Score
SSFE 3 + CA + ARVFL (without DRA)	87.12	87.36	87.11	87.12	85.99
SSFE-3 + DRA + ARVFL (without CA)	94.15	94.43	94.02	94.15	93.9
MSFE-35 + Dual Att + ARVFL (with multi-scale features)	87.23	87.59	87.10	87.23	86.08
SSFE-3 + Dense + Dual Att (without RVFL Attention)	88.61	88.37	88.22	88.61	87.79
Proposed Study	98.83	99.05	97.42	98.82	98.21

SSFE 3: Single Stage Feature (kernel size: 3) Extractor; DRA: DR Attention; CA: Cross Attention; ARVFL: Adaptive RVFL; MSFE 35: Multi-Stage Feature (kernel size: 3 & 5), Dual Att: Dual Attention.

TABLE 8. Comparative Summary of performance (%) obtained by comparing with pretrained backbone models.

Models	Accuracy	Precision	Specificity	Sensitivity	F1 Score
MobileNet V2	96.27	93.42	97.36	95.30	94.33
Efficient Net B0	97.55	97.57	97.23	97.55	97.56
Efficient Net B1	92.76	95.81	91.18	81.13	86.82
NASNet Mobile	94.14	92.74	94.09	87.58	89.98
DenseNet 121	88.72	89.62	88.82	88.72	88.99
ConvNeXtBase	88.81	89.26	89.72	88.14	88.38
InceptionResNetV2	98.2	96.72	96.98	97.65	96.64
ViT	93.12	96.05	90.83	81.81	87.19
Proposed Study	98.83	99.05	97.42	98.82	98.21

This suggests that the inclusion of data from multiple centers enhances the stability and robustness of feature attributions, mitigating potential biases introduced by localized data characteristics. For the Figure 7 b) exhibits greater variability in feature contributions, with some features exerting highly positive effects. This reflects a higher degree of heterogeneity and potential over-reliance on specific features, likely due to the localized nature of the data.

V. DISCUSSION AND FUTURE WORKS

A. EXPLAINABILITY AND UNCERTAINTY QUANTIFICATION

In medical imaging, particularly for brain ischemic stroke studies, the application of ML and DL has significantly transformed diagnostic processes. However, the complexity and opacity of these algorithms present challenges regarding their interpretability and explainability, which are crucial for clinical acceptance and decision-making. The traditional ''black box'' nature of DL models presents a significant challenge to clinical acceptance [85]. To address this, XAI tools, such as saliency maps, have been developed to visualize the decision-making process. Radiomics further advances this by transforming subjective visual assessment into objective interpretation through quantitative features including shape, intensity, and texture measurements [86].

Uncertainty quantification introduces another crucial dimension to model reliability. Aleatoric uncertainty captures inherent variability in imaging data, accounting for noise, acquisition variations, and image quality differences. This becomes particularly relevant in multi-center validations where imaging protocols may vary. Epistemic uncertainty, on the other hand, reflects the model's confidence in its predictions, proving especially valuable for borderline cases between acute and chronic classifications where additional clinical context may be necessary [85], [86].

B. INSIGHTS ON THE MODULES OF THE PROPOSED MODEL

The Multi-Scale Feature Extraction Module employs parallel convolutional pathways with progressive channel expansion *(32 → 64 → 128 → 256)* and interspersed max-pooling operations. This architecture enables simultaneous processing of fine-grained details and broader contextual information, crucial for capturing stroke-related anomalies while remaining robust to variations in image scale and complexity.

The DA mechanism synergizes DR and CA, leveraging complementary strengths to improve feature discrimination. DR captures the structural organization of features (Hierarchical Feature Learning), essential for distinguishing ischemic regions from normal tissues. CA facilitates inter-sequence relationship modelling by aligning query and key representations, hence enabling it to learn Context-Aware Representations (dependencies across spatial and semantic dimensions), vital for differentiating subtle ischemic patterns. The ARVFL layer enhances the traditional RVFL network by introducing an adaptive factor that adjusts the initially randomized weights based on the input data. Unlike the static weights in standard RVFL, this layer learns trainable to modulate the weights dynamically, improving the model's ability to capture complex patterns. By combining these adaptive weights with direct input-output connections and bias terms, the layer retains the simplicity and computational efficiency of RVFL while significantly improving its flexibility and learning capacity. It's faster training process as it eliminates the need of back propagation, the direct link from input-output links accelerates the convergence. The time-varying weights enable the model to dynamically adjust to data variability, ensuring robustness in dynamic environments such as real-time stroke detection. For brain CT image diagnosis, this adaptability is particularly advantageous. Brain CT images often exhibit subtle and variable patterns, such as ischemic changes, that require nuanced feature extraction. The adaptive mechanism allows the layer to tailor its representation to these variations, enhancing its sensitivity to key diagnostic features while maintaining low computational overhead. This makes it well-suited

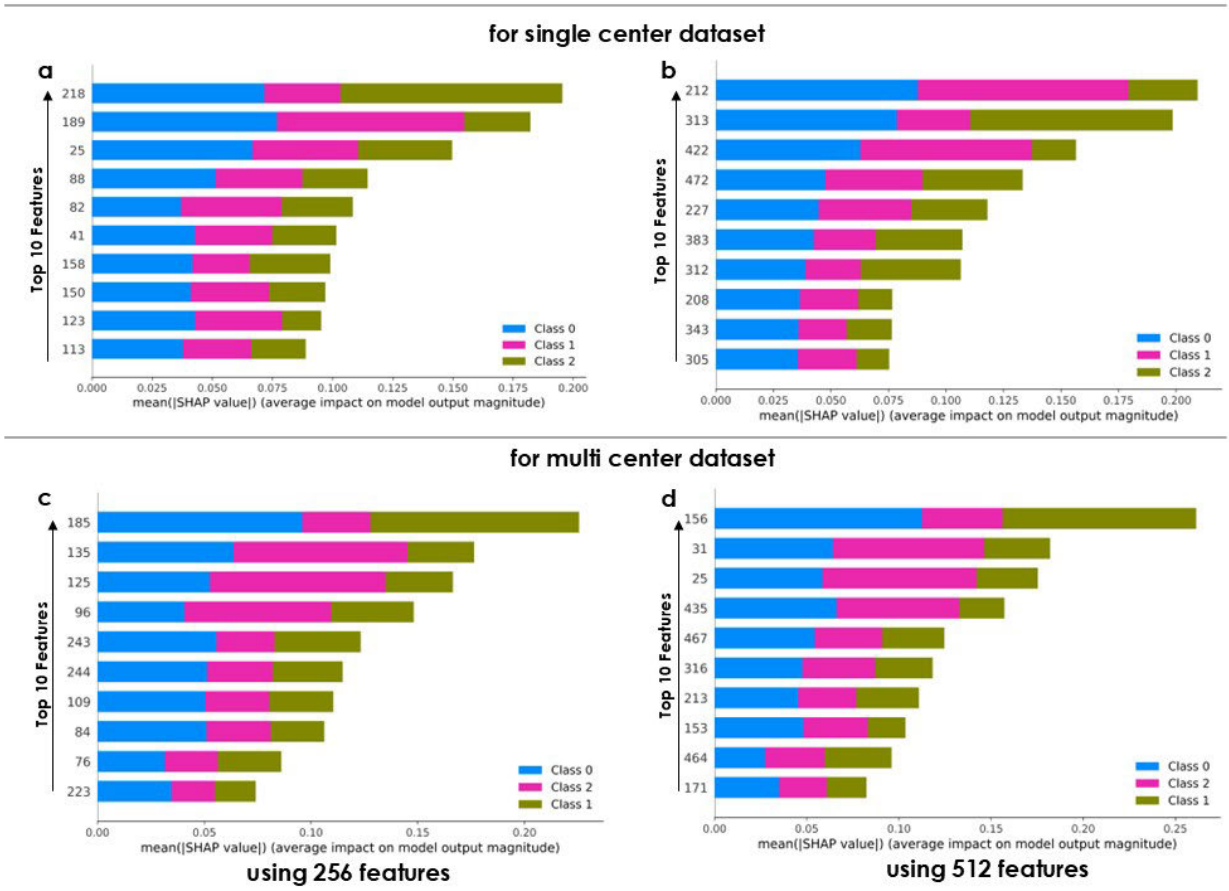


FIGURE 8. Summary plot of mean SHAP values across classes.

for real-time applications and resource-constrained environments, such as mobile or bedside diagnostic systems. In terms time complexity of the proposed approach integrates three major components, *i.e.,* Multi-Scale Feature Extraction (MSFE) module utilizes parallel P convolutional pathways with varying kernel sizes and progressive channel dimensions is given by:

$$T_{\text{MSFE}} = P \cdot \sum_{l=1}^{L} O\left(K_l^2 \cdot C_{l,\text{in}} \cdot C_{l,\text{out}} \cdot Height_l \cdot Width_l\right) \tag{13}$$

where

- K is the kernel size,
- C_{in} and C_{out} are the number of input and output channels respectively,
- *Height* and *Width* are the height and width of the feature map, and
- L is the number of layers per pathway, K_l varies across kernels *3,5, 3&5*.

In the case of the DA mechanism, DR involves the iterative computation of coupling coefficients and weighted sums. Therefore, the time complexity for DR is:

$$T_{\text{DR}} = O(I \cdot N \cdot C_{\text{in}} \cdot C_{\text{out}}) \tag{14}$$

where

- I is the number of routing iterations, and
- N is the number of capsules.

CA involves matrix multiplication and softmax operations, so the time complexity for CA is:

$$T_{\text{CA}} = O(\hat{n_q} \cdot \hat{n_r} \cdot \hat{n_v}) \tag{15}$$

where

- $\hat{n_q}$, $\hat{n_r}$, and $\hat{n_v}$ are the sizes of the query, key, and value matrices.

Lastly, the ARVFL layer involves matrix multiplication for input projection and output computation. The time complexity for the ARVFL layer is:

$$T_{\text{ARVFL}} = O(C_{\text{in}} \cdot C_{\text{hidden}} + C_{\text{hidden}} \cdot C_{\text{out}}) \tag{16}$$

where

- C_{hidden} is the number of hidden units.

To analyze the separability and structure of features extracted by the ARVFL layer, we employed t-distributed Stochastic

TABLE 9. Comparison of automated multi-class brain stroke classification methods using CT images.

Authors	Class	Dataset type	Methodology	Results
Qasrawi et al. [21]	4: Hyperacute, Acute, Sub-acute, and Chronic	single-center	Hybrid ensemble with DenseNet121 and edRVFL	Acc.: 93.3% (for hyper-acute); Acc.: 94.8% (for acute); Acc.: 97.4% (for sub-acute); Acc.: 98.2% (for chronic)
Gautham et al. [23]	3: Hemorrhagic, Ischemic, and Normal	single-center	Gradient pattern with fine KNN	Acc.: 86.11%
Gautham et al. [24]	3: Hemorrhagic, Ischemic, and Normal	single-center	Local neighbourhood pattern with ensemble bagged trees	Acc: 82.65%
Neethi et al. [36]	3: Hemorrhagic, Ischemic, and Normal	single-center	3D CNN	Acc.: 92%
Raj et al. [44]	3: Haemorrhage, Infract, and Normal	single-center	CNN, ViT & AutoML	Acc.: 92%
Hossain et al. [49]	3: Hemorrhagic, Ischemic, and Normal	single-center	ViT + LSTM	Acc.: 94.55% (on BrSCTHD 2023 dataset); Acc.: 96.61% on Kaggle dataset
Proposed	**3: Acute, Chronic, and Normal**	**single-center and multi-center**	**Multi-scale feature extraction with dual attention and Adaptive RVFL for classification**	**Max. Acc: 98.83% (for single-center); Max. Acc: 92.42% (for multi-center)**

Neighbor Embedding (t-SNE). The ARVFL-extracted features, with dimensions of *256* and *512*, were reduced to two dimensions and plotted for both single-center and multi-center datasets. These plots, presented in Figure 9, provide insights into the discriminative capability of the extracted features and their adaptability to varied data distributions.

For the single-center dataset, the t-SNE visualization reveals well-clustered and distinct groupings corresponding to the three classes: acute, chronic, and normal. This clear separation indicates that the ARVFL layer effectively projects the input data into a feature space where classes are well-separated, facilitating accurate classification by the LS algorithm. The transition from *256* to *512* features enhances the granularity of clusters, suggesting that higher-dimensional representations from the ARVFL layer capture more nuanced patterns in the data, further improving class separation. In contrast, the t-SNE plots for the multi-center dataset exhibit more overlapping regions, reflecting the inherent variability and distributional shifts present in data collected from multiple centers. However, despite this increased complexity, the clusters remain distinguishable, particularly with the *512*-dimensional features, which demonstrate better separation compared to the *256*-dimensional counterparts. This underscores the robustness of the ARVFL layer in adapting to heterogeneous data distributions. The ability of the ARVFL layer to generate features that retain discriminative information across diverse clinical settings validates its effectiveness in enabling the LS algorithm to perform well, even under challenging multi-center conditions.

C. COMPARATIVE RESULTS

Table 9 shows the summary of comparative studies done on automated brain stroke classification for multi-class using CT images. It is noted that proposed approach has shown promising results for both single and multi-center data. In this section, we have compared recently published studies for multi-class brain stroke classification employing state-of -the-art techniques. Neethi et al. [36] and Raj et al. [44] employed 3D CNN and ViT-based models for three-class classification of brain stroke (*i.e.,* hemorrhagic, ischemic/infracts, and normal), achieving 92% accuracy but lacking attention mechanisms for fine-grained feature extraction. Gautham et al. [23], [24] have used gradient-based with fine KNN and local neighborhood pattern with ensemble bagged trees and achieved accuracies of 86.11% and 86.69% respectively, however their reliance on handcrafted features limits adaptability. Moreover, the four-class problem (*i.e.,* hyperacute, acute, sub-acute, and chronic) was tackled by Qasrawi et al. [21] using a DenseNet121 and edRVFL ensemble model and achieved accuracies of 93.3% to 98.2% with high computational cost. Recently, Hossain et al. [49] have used the ViT-LSTM hybrid model and achieved 96.61% accuracy, but lacked the region-specific feature discrimination ability. It is noted that an exact comparison

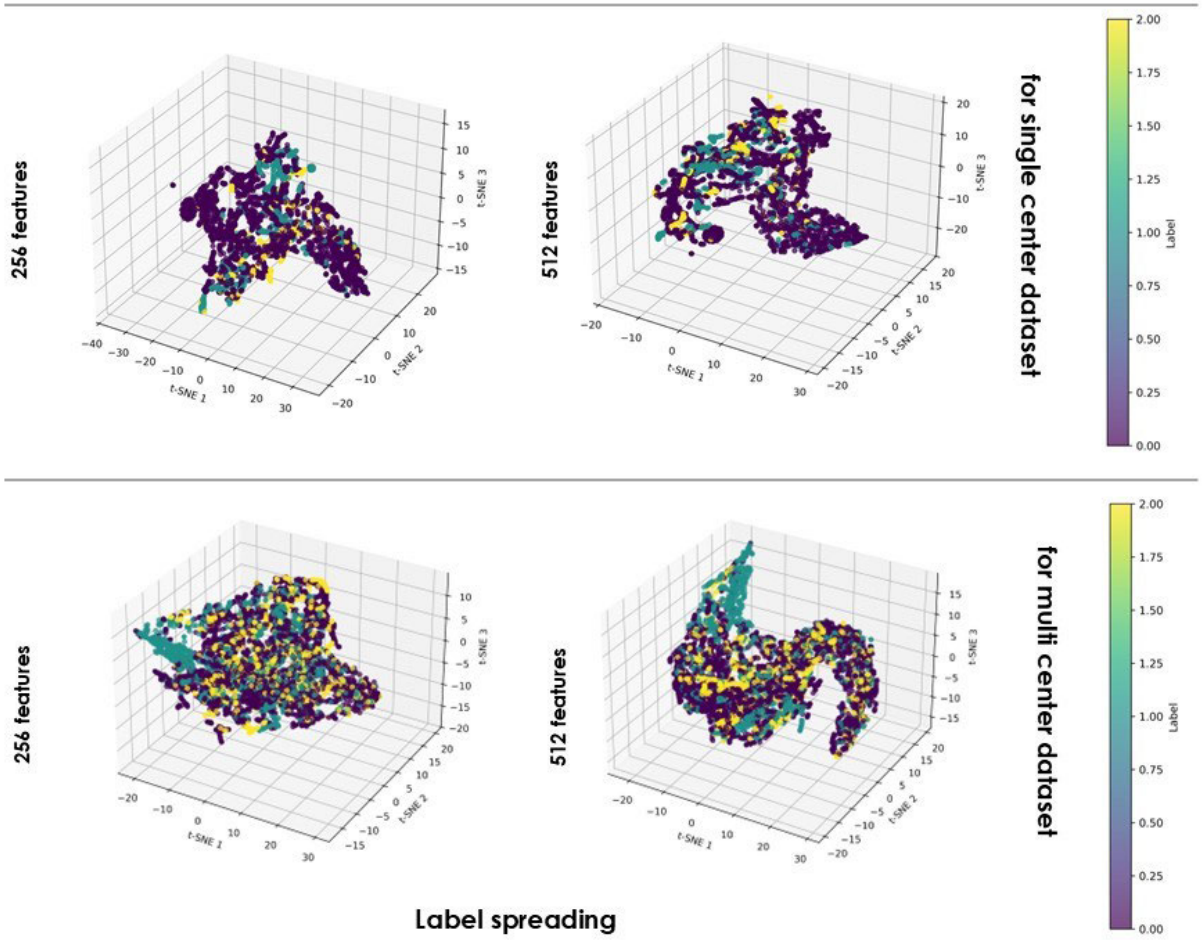


FIGURE 9. t-SNE distribution plot for the single and multi-centre data with 256 and 512 features.

cannot be made amongst all these studies, as these methods are experimented using private datasets. However, in terms of generalization over multi-centric data our proposed model performed best and yielded promising results *i.e.,* increased accuracy (98.83%), reduced false positives (precision of 99.05%) and false negatives (sensitivity of 97.42%).

D. CHALLENGES IN BRAIN STROKE DETECTION AND CLASSIFICATION

The detection and classification of ischemic brain stroke based on CT images face significant obstacles that affect the efficiency and accuracy of ML and DL model. One of the biggest problems is related to the quality of data used to train these models. CT image datasets often suffer from institutional variations in imaging protocols, storage methodologies, and data sharing restrictions [85]. These inconsistencies limit the development of robust, generalizable models across different clinical settings. The need for extensive, high-quality, systematically organized datasets remains a fundamental challenge. The complexity involved in the algorithms of ML and DL poses an additional challenge. Most of these models, especially the ones related to DL, work as ''black boxes,'' which makes it difficult to interpret their decision-making process. This lack of transparency creates a problem when the clinicians are asked to explain such decisions to the patients or to implement them in the clinical practice [85]. The formation of XAI tools, which includes saliency maps, is necessary for promoting trust and usability in clinical environments [85]. This is a typical issue with ML models wherein the model learns to fit the noise present in the training data rather than the actual information within the data. Consequently, the model performs poorly on unseen data [85]. Among reducing overfitting and achieving the utmost generalizability, regularization, batch normalization, as well as the usage of larger training datasets are needed [85]. Nevertheless, the difficulty persists in obtaining enough representative data to adequately train and validate these models. Moreover, the uneven distribution of class labels in stroke cases complicates this evaluation, as traditional metrics may not capture the true performance of the model across all subtypes [87]. The integration of automated stroke detection systems into existing clinical workflows poses several logistical challenges. These systems must be validated in real-world settings and operate seamlessly with current

diagnosis techniques to ensure that they enhance, not detract from, patient care [88]. In addition, greater validation is needed on larger and more heterogeneous datasets to confirm the robustness of the models and their ability to handle variability in patient data [85].

E. FUTURE DIRECTIONS

Future research in ischemic stroke detection will benefit from diverse data integration, combining genetic markers, lifestyle factors, and advanced imaging [88], [89]. Novel DL architectures focusing on clinical interpretability, alongside advances in ensemble learning methods, promise improved diagnostic [88]. This multi-modal approach aims to identify high-risk patients for timely intervention. To ensure the practical applicability of such models, a multi-phase clinical validation approach is needed. Such an approach should starts with a pilot observational study in multiple hospitals, involving considerable number of patients to assess how accurately the model can predict the diagnosis compared with established diagnostics [88]. Thorough clinical validation seeks to enhance the reliability and impact power of the model on patient care, addressing the deficiencies of previous studies that relied mostly on small sample sizes in many cases [90]. Collaboration with medical fraternity and financial institutions would be sought for this purposeful endeavor [88]. The integration of various imaging techniques represents one of the most important parts of the future landscape of stroke diagnosis. While NCCT is currently the first choice for ruling out intracranial hemorrhage, better cost-effective and accessible diagnostic tools will be important determinants of quicker treatment times [89], [91]. Innovations in imaging, especially MRI-based ones, given its sensitivity, may assist in the better and earlier diagnosis and assessment of acute ischemic strokes despite the constraints on availability of resources [91]. There is also inherent weakness in ML algorithms, to which model over-fit and under-fit contributed [90]. This may reduce the usability in clinical environments. To get better robustness in the model and less false positives, this calls for high standards in the pre-processing of images and proper feature determination [90]. Further, validation of algorithms outside hospitals is necessary for establishing their general applicability in different clinical contexts, thus allowing healthcare practitioners to rely on and effectively incorporate these instruments into their decision-making frameworks [88], [90].

VI. CONCLUSION

This study introduces an innovative DL framework for ischemic stroke classification in CT images, advancing the state-of-the-art through several significant contributions. Our framework combines Multi-Scale Feature Extraction with Dual Attention mechanisms to detect subtle radiological markers often missed by conventional methods. The integration of an ARVFL layer significantly improves computational efficiency while maintaining high classification performance. Validation on a multi-center dataset demonstrates the model's robustness and generalizability, achieving maximum 98.83% accuracy in classifying normal and acute, chronic stroke cases. Additionally, the incorporation of XAI tools bridges the gap between technical performance and clinical usability, fostering trust and transparency in diagnostic decision-making. The proposed framework not only outperforms existing state-of-the-art models but also offers scalability for deployment in resource-constrained settings, making it a practical solution for real-world clinical applications. Future work will focus on extending the model's capabilities through the integration of additional data modalities and real-time clinical validation studies. These enhancements aim to further solidify its role as a reliable and efficient tool for improving patient outcomes in time-critical stroke management scenarios.

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MAHESH ANIL INAMDAR is currently pursuing the Ph.D. degree in the domain of prospects of artificial intelligence in health care (ischemic brain strokes). He is a Faculty Member with the Department of Mechatronics. He has industrial experience as a Data Scientist in the machine learning work domain. His research interest includes the development of intelligence systems that can improve and revolutionize the current healthcare systems. More information can be found at: https://scholar.google.com/citations?user=Aru-8wQAAAAJ&hl=en.



ANJAN GUDIGAR received the Ph.D. degree from Manipal Academy of Higher Education, India. He is currently a Faculty Member with the Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal, India, and an Adjunct Associate Professor with the School of Medicine, Taylor's University, Malaysia. He has published several research papers in international conferences and journals. His research interests include image processing, medical image analysis, and pattern recognition. For more information visit the link (http://scholar.google.co.in/citations?user=qoe6EvsAAAAJ&hl=en).



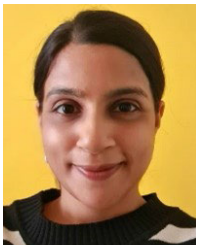
U. RAGHAVENDRA received the Ph.D. degree from Manipal Academy of Higher Education, India. He is currently a Faculty Member with the Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal, India. He has published several papers in refereed international SCI-IF journals and international conference proceedings. He has a patent to his credit and received an Invention Award from Intellectual Ventures, USA, for his innovations, in 2014. His major research interests include 3D computer vision, image processing, and medical image analysis. More information can be found at: https://scholar.google.co.in/citations?user= 3nzcDREAAAAJ&hl=en.



MASSIMO SALVI received the master's degree in biomedical engineering from the Politecnico di Torino, Turin, Italy, in 2014, and the Ph.D. degree, in 2018, with a thesis titled "Modeling and Interpretation of Physiopathological Processes using Multimodal and Multiscale Medical Imaging." In addition to his work in biomedical signal and image processing, he is also actively engaged in the field of artificial intelligence and deep learning. He has experience in the development of hybrid techniques that combine AI-based methods and mathematical-statistical techniques to address complex problems in biomedical research. His research is primarily focused on the development of automated solutions for fluorescence and optical microscopy, with a particular emphasis on the analysis and interpretation of cellular and tissue images.



RAJA RIZAL AZMAN BIN RAJA AMAN received the medical degree from King's College London, in 2002. He trained as a Radiologist with Universiti Malaya, Kuala Lumpur, Malaysia, and as a Neuroradiologist with The University of Hong Kong, China. He was an Associate Professor of radiology with Universiti Malaya. He currently practices at the Sunway Medical Centre, Selangor, Malaysia. He is a member of the Royal College of Physicians (MRCP) and a fellow of the Royal College of Radiologists (FRCR).



NADIA FAREEDA MUHAMMAD GOWDH received the M.B.B.S. and Master of Radiology degrees from Universiti Malaya, Malaysia, in 2006 and 2017, respectively. She is currently a Clinical and Academic Radiologist with the Biomedical Imaging Department, Faculty of Medicine, Universiti Malaya. She is a Practicing Radiologist and has an active role as a Principal Investigator and a Co-Investigator in several medical imaging-related research projects. Her research interests include computed tomography, magnetic resonance imaging, neuroradiology, and pediatric radiology, and is aimed at improving patient care through the application of these imaging technologies. In addition to that, she is also involved in academic duties, particularly in the teaching and supervision of post-graduate students in the radiology program.



IZZAH AMIRAH BINTI MOHD AHIR received the Bachelor of Medicine, Bachelor of Surgery (M.B.B.S.) degree from Universiti Malaya, Malaysia, in 2014, where she is currently pursuing the Master of Radiology degree. She is a Medical Officer with the Department of Biomedical Imaging, University of Malaya Medical Center (UMMC). Before joining UMMC, she completed two years of housemanship training at Melaka Hospital and worked as a Medical Officer at Kuala Lumpur Hospital and Shah Alam Hospital. She is conducting a thesis research project focused on the outcomes of treatment for large vessel occlusion at UMMC.



MOHD SALAHUDDIN BIN KAMARUDDIN received the Bachelor of Medicine, Bachelor of Surgery (M.B.B.S.) and Master of Radiology (MRAD) degrees from Universiti Malaya. He is currently a Clinical Radiologist and a Medical Lecturer with the Department of Biomedical Imaging, Faculty of Medicine, Universiti Malaya. His clinical focus includes neuroradiology and interventional radiology with a particular interest in advancing expertise in neurointerventional radiology.



KHAIRUL AZMI ABDUL KADIR received the M.B.B.S. and Master of Radiology degrees from Universiti Malaya, Malaysia, in 1997 and 2004, respectively. He obtained his neurosurgery endovascular training at Hacettepe University, Türkiye. He is currently a Neurointerventional Radiologist with the Biomedical Imaging Department, Faculty of Medicine, Universiti Malaya. He has extensive experience in stroke imaging and neurointervention, playing an active role in both clinical practice and research. His research interests include advancing stroke imaging and intervention techniques reflecting his keen interest in improving patient outcomes in these critical areas. In addition to his clinical and research roles, he contributes actively to hospital administration, formulation of national guidelines, and provides international consultation on neurointervention-related topics.



FILIPPO MOLINARI (Senior Member, IEEE) is currently a Professor of biomedical engineering with the Faculty of the Department of Electronics and Telecommunications, Politecnico di Torino, Turin, Italy. His current research interests include medical imaging, ultrasound technologies, and the noninvasive assessment of cerebral functions and autoregulation. He is the Editor-in-Chief of *Computer Methods and Programs in Biomedicine*. More information can be found at: https://scholar.google.it/citations?user=ttbUYiQAAAAJ&hl=it.



AJAY HEGDE is an experienced and highly qualified Neurosurgeon in Bengaluru. As a dedicated Surgeon, he completed his training in neurosurgery at the prestigious Kasturba Medical College, Manipal. He pursued further training in functional neurosurgery and neurooncology at the Institute of Neurological Sciences, Glasgow, U.K., and secured the Fellowship of the Royal College of Surgeons of Edinburgh (FRCS). More information can be found at: https://scholar.google.com/citations?user=B8uP4dwAAAAJ&hl=en&oi=ao.



GIRISH R. MENON is currently a Professor of neurosurgery with extensive experience in academic and clinical settings, currently heading the Neurosurgery Department, Kasturba Medical College, Manipal. He is recognized for establishing neurosurgical departments in multiple institutions, including hospitals in South Africa and India. His expertise spans cerebrovascular surgery, epilepsy surgery, and pediatric neurosurgery, with a track record of expanding surgical programs and academic training. He conducted multiple research projects with both internal and external funding, resulting in over 169 publications and an H-index of 18. He served on the editorial boards and delivered numerous invited talks at national and international conferences. More information can be found at: https://scholar.google.com/citations?user=vXSrrVwAAAAJ&hl=en.



U. RAJENDRA ACHARYA received the D.Sc., D.Eng., and Ph.D. degrees. He is currently a Professor with the School of Mathematics, Physics and Computing, University of Southern Queensland, Australia; a Distinguished Professor with the International Research Organization for Advanced Science and Technology, Kumamoto University, Japan; an Adjunct Professor with the University of Malaya, Malaysia; and an Adjunct Professor with Asia University, Taiwan. His funded research has accrued cumulative grants exceeding six million Singapore dollars. He has authored over 800 publications, including 750 in refereed international journals, 42 in international conference proceedings, and 17 books. He has received over 75,000 citations on Google Scholar, with an H-index of 137. His research interests include biomedical imaging and signal processing, data mining, and visualization, as well as applications of biophysics for better healthcare design and delivery. He sits on the editorial boards of multiple journals and has served as the guest editor on several AI-related issues. According to the Essential Science Indicators by Thomson, he consistently ranked among the top 1% of Highly Cited Researchers in Computer Science for the last eight years, from 2016 to 2023. More information can be found at: https://scholar.google.com/citations?hl=en&user=8FjY99sAAAAJ.